Jenna M. Kline

Github | LinkedIn | Google Scholar | jennamk9@mit.edu

Education

PhD Computer Science & Engineering, **The Ohio State University** 2021 — 2026

Dissertation: Autonomous Drone Systems for In Situ Animal Ecology: Towards AI-Driven Multi-Modal Ecosystem Monitoring at the Far Edge
Major: Software Systems | ***Minors:*** Artificial Intelligence & Imageomics
Advisors: Dr. Christopher Stewart and Dr. Tanya Berger-Wolf

B.S. Industrial and Systems Engineering, *cum laude*, **The Ohio State University** 2014 — 2018

Honors in Integrated Business and Engineering (IBE)

Research Interests

Cyber-Physical Systems for Adaptive Environmental Sensing at the Far Edge

I build **cyber-physical systems** for autonomous environmental sensing in dynamic, resource-constrained field settings. My research advances **distributed intelligence**, **edge AI**, and **adaptive autonomy** to coordinate heterogeneous sensors and robots, enabling **cross-modal systems** to adapt what, where, and when they observe. I deploy these systems across **ecology**, **biodiversity monitoring**, and **coastal and climate resilience**, developing the foundations for persistent, adaptive observation of complex environmental systems.

Academic Appointments and Experience

Massachusetts Institute of Technology, Postdoctoral Associate Jun 2026 — present

Department of Civil and Environmental Engineering

Research on coastal resilience, using satellite imagery, drones and computer vision to monitor salt marsh habitats at Cape Cod through the Climate Project at MIT, supervised by Dr. Heidi Nepf.

The Ohio State University, Graduate Research Assistant Aug 2021 — May 2026

Department of Computer Science and Engineering

Conducted research on autonomous drone systems for in situ animal ecology, including multi-modal sensing, cross-modal coordination, and edge AI. Research supported by Ohio State University Presidential Fellowship and NSF Imageomics (Award 2118240) and ICICLE (Award 2112606).

Visiting PhD Candidate, WildDroneEU 2024 — 2026

Conducted research in collaboration with the WildDrone consortium. Completed two research visits, at the University of Southern Denmark (2024) and the University of Bristol (2024), and presented at the WildDrone summer school hosted by the Max Planck Institute of Animal Behavior, Konstanz (2025). Joined two drone field campaigns at Ol Pejeta Conservancy, Kenya (Jan 2025; Feb–Mar 2026 WildDrone Hackathon), testing the WildWing and behavior-adaptive drone systems [F5, F4] while supporting collaborators' flight operations and data collection.

Los Alamos National Laboratory, Supercomputer Institute Graduate Intern Summer 2023

Completed high performance computing bootcamp. Conducted research on scalable characterization of HPC filesystem trees, presented at SC23 ACM Student Research Competition.

Skills

Field Research and Leadership: FAA Part 107 Remote Pilot Certificate (small UAS) · drone piloting and flight operations · field campaign planning, training,and coordination (Kenya, Ohio, Massachusetts)

Autonomous Sensing: autonomous flight control · mission planning · perception–control integration · adaptive navigation policies · multimodal sensor networks (drone, camera trap, ARU)

Edge AI and Distributed Computing: real-time AI inference on resource-constrained hardware · edge and embedded platforms · hardware-aware system design · edge–cloud orchestration · workload and performance characterization · shared HPC environments · distributed storage and filesystems · wildlife detection, tracking, pose estimation, behavior recognition, and re-identification with AI, including edge inference on drones and embedded devices

Data Infrastructure: FAIR and AI-ready metadata standards · multimodal dataset design and curation · spatiotemporal data synchronization · annotation workflows (GitHub) · reproducible data releases and benchmarking (Hugging Face, Zenodo)

Programming and Tools: Python · PyTorch · SQL · Linux · Git · scientific computing and data-analysis workflows

Featured Publications

*Complete record in Research Outputs (p. 4): F featured, J journal, C conference, P presentation, A artifact; * student mentee.* Note: Computer science conference publications are archival full papers that undergo rigorous peer review, as is typical for journal publications. Venue acceptance rate (**AR**) and impact factor (**JIF**) reported where available.

[F1] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline, K. Meier, V. Shukla, E. G. A. Rolland, E. Iannino, et al.
Manuscript under review, **2026**. *Journal paper.*

[F2] **A Blueprint for Cross-Modal Coordination at the Far Edge**
J. Kline, A. Potlapally*, H. Subramoni, T. Berger-Wolf, and C. Stewart
ACM/IEEE Symposium on Edge Computing (SEC), **2026**. *Conference full paper (to appear).*

[F3] **Studying Collective Animal Behaviour with Drones and Computer Vision**
J. Kline, S. Afridi, E. G. A. Rolland, G. Maalouf, L. Laporte-Devylder, et al.
Methods in Ecology and Evolution 16(10), **2025**. *Journal paper.* AR **25%**, JIF **6.2**.
Robert May Prize for best paper from early-career author.

[F4] **Edge-Native, Behavior-Adaptive Drone System for Wildlife Monitoring**
J. Kline, R. Katole, T. Berger-Wolf, and C. Stewart
DroneSys: Workshop on Autonomous Drone Computing Systems and Application at the ACM/IEEE Symposium on Edge Computing (SEC), **2025**. *Conference workshop paper.*
Associated work awarded **First Place in the Student Research Competition (SRC).**

[F5] **WildWing: An Open-Source, Autonomous and Affordable UAS for Animal Behaviour Video Monitoring**
J. Kline, A. Zhong*, K. Irizarry, C. V. Stewart, C. Stewart, et al.
Methods in Ecology and Evolution 17(2), **2025**. *Journal paper.* AR **25%**, JIF **6.2**.

[F6] **Characterizing and Modeling AI-Driven Animal Ecology Studies at the Edge**
J. Kline, A. O'Quinn, T. Berger-Wolf, and C. Stewart
ACM/IEEE Symposium on Edge Computing (SEC), **2024**. *Conference full paper.* **27%** AR.

[F7] **A Framework for Autonomic Computing for In Situ Imageomics**
J. Kline, C. Stewart, T. Berger-Wolf, M. Ramirez, S. Stevens, et al.
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), **2023**. *Conference vision paper.* **23%** AR.

Grants and Awards

Presidential Fellow: The Ohio State University Presidential Fellowship	2025 — 2026
Robert May Prize: Best early-career-author paper, British Ecological Society [F3]	2025
Student Research Competition - First Place: ACM/IEEE Symposium on Edge Computing [P12]	2025
Departmental Research Award: Outstanding Graduate Research in CSE, OSU	2025
Hayes Forum - First Place: Hayes Advanced Research Forum, OSU [P21]	2025
Rising Stars in EECS: MIT & Boston University Rising Stars Workshop	2025
WiscProf Workshop: Future Faculty in Engineering, Univ. of Wisconsin–Madison	2025
Research Grant: OSU Alumni Grant for Graduate Research and Scholarship	2025
NSF NAIRR Pilot Allocation: Digital Twins for Ecological Research (NAIRR250091)	2025
Best Post Poster Winner: Autonomic Computing & Self-Organizing Systems (ACSOS) [C14]	2024
TDAI Best Poster Award: OSU Translational Data Analytics Institute Fall Forum [P22]	2024
Travel Grant: ACM/IEEE Symposium on Edge Computing	2022 & 2024
CRA-WP Grad Cohort: Computing Research Association Grad Cohort for Women [P24]	2023
Grace Hopper Scholar: The Ohio State University ACM-W	2022

Field Experience

Waquoit Bay National Estuarine Research Reserve, Massachusetts Summer 2026

Team lead and drone pilot. Conducted drone fieldwork monitoring salt-marsh habitats with computer vision, supervising a team of undergraduate researchers. Coastal monitoring project at with Dr. Heidi Nepf, part of MIT's Climate Project.

Ol Pejeta Conservancy, Kenya Feb — Mar 2026

Drone pilot. Tested behavior-adaptive drone system for wildlife monitoring [F4]. Funded by the Ohio State University Graduate School's Alumni Grants for Graduate Research and Scholarship (AGGRS).

The Wilds Conservation Center, Ohio Summer 2025

Team lead. Collected the SmartWilds multimodal dataset — camera-trap, bioacoustic, and drone data [C7].

Ol Pejeta Conservancy, Kenya Jan 2025

Drone pilot. Tested the WildWing system [F5] and collected the MMLA dataset [C9].

The Wilds Conservation Center, Ohio Summer — Fall 2024

Lead drone pilot. Tested the WildWing system [F5] and collected the MMLA dataset [C9].

Mpala Research Centre, Kenya Jan 2023

Lead drone pilot. Collected the KABR dataset for in situ animal behavior recognition [C12].

Mentoring and Teaching

Mentoring and Student Supervision

Massachusetts Institute of Technology (2026–present). Supervised two summer undergraduate researchers on a salt-marsh monitoring study using drones and computer vision.

The Ohio State University (2022–2026). Supervised five undergraduate researchers, including a team conducting fieldwork at The Wilds Conservation Center, and led a field-campaign team of undergraduate, master's, and PhD students. Student-led projects were published at CV4Animals at CVPR 2026 [C4, C5] and SEC 2025 [P13].

University of Bologna (2024). Co-supervised Denys Grushchak's master's thesis, *Herd Monitoring with Autonomous Drones: A Decentralized k-Coverage-Inspired Approach*, with Dr. Danilo Pianini; the thesis resulted in two publications at the IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS) [C14, C15].

Teaching Experience

All courses taught at The Ohio State University, Department of Computer Science and Engineering.

Lab Instructor — *CSE 2111: Modeling and Problem Solving with Spreadsheets and Databases.* Undergraduate required course. Led two 30-student lab sections twice weekly and supervised a team of undergraduate graders. Topics: spreadsheet and database modeling, programming concepts, business applications.

Graduate Teaching Assistant — *CSE 3244: Data Management in the Cloud.* Senior-level elective. Sole graduate teaching assistant responsible for grading and supervising lab assignments and the final project. Topics: data organization on cloud computing architectures, B-tree and hash-based indexing, query optimization fundamentals, data partitioning, and distributed task scheduling.

Service

Program Committee Member: AI4Nature Workshop on Computer Vision and AI for Environmental Monitoring, Biodiversity Analysis, and Ecosystem Restoration, IEEE AVSS 2026

Community Building Co-chair: ACSOS 2026; organized community-building seminar series 2026 — present

Workshop Organizer & Lead Facilitator: Edge AI for Conservation workshop, International Conservation on Technology for Conference (ICTC), Lima, Peru 2026

Group Co-lead: Edge Computing for Conservation Group, WILDLABS Network 2025 — present

Workshop Organizer: Imageomics Image Datapalooza Hackathon 2023

Volunteer: Code I/O coding camp, The Ohio State University ACM-W Chapter 2023

Peer Mentor: Graduate Student Peer Mentoring Program, The Ohio State University 2022 — 2024

Journal Reviewer: International Journal of Computer Vision; Methods in Ecology and Evolution; Scientific Reports; Ecological Informatics; Ecological Modelling; Global Ecology and Conservation; Marine Mammal Science; PLOS ONE; IET Computer Vision; Ecology and Evolution; MethodsX.

Research Outputs

Given the interdisciplinary and collaborative nature of my work, this section lists both computer science conference papers and journal publications, as well as associated preprints. Computer science conference publications (SEC, EDGE, ACSOS, etc.) are archival full papers that undergo rigorous peer review, as is typical for journal publications. Accepted ecology conference abstracts are listed under Presentations. Venue acceptance rate (**AR**) and impact factor (**JIF**) reported where available.

Journal Articles

[J1] **When to Trust Automation for In Situ Animal Behavior Monitoring: An Empirical Evaluation from Drone and Camera-Trap Video**
J. Kline, B. Braasch, B. Fischer, A. Zhong*, M. Thompson, et al.
Manuscript in preparation, **2026**.

[J2] **Putting the "Adaptive" in Adaptive Monitoring: From Fast Data to Meaningful Ecological Change**
L. J. Pollock, P. H. P. Braga, C. R. Florian, K. Hébert, **J. Kline**, et al.
Manuscript under review; EcoEvoRxiv preprint, **2026**.

[J3] **Simulated and Field-Based Error Characterization of Animal Geolocalisation and Relative Positioning via Commercial Drones**
K. Meier, E. G. A. Rolland, E. Iannino, M. Simpson, **J. Kline**, et al.
Manuscript under review, **2026**.

[J4] **Simulation-Grounded Decentralized Multi-UAV Coordination for Scalable In-Situ Imageomics of Wildlife Herds**
G. Aguzzi, M. Baiardi, N. Farabegoli, D. Grushchak*, **J. Kline**, D. Pianini, C. Stewart, and M. Viroli
Manuscript under review, **2026**.

[J5] **SORA 2.5-Guided BVLOS UAS for Wildlife Conservation in Kenya: Reducing Friction Between Safety and Field Operations**
G. Maalouf et al., including **J. Kline**
Drones 10(3), article 178, **2026**. JIF **5.2**.

[J6] **BaboonLand Dataset: Tracking Primates in the Wild and Automating Behaviour Recognition from Drone Videos**
I. Duporge, M. Kholiavchenko, et al., including **J. Kline**
International Journal of Computer Vision 133(9), **2025**. JIF **10.3**.

[J7] **Deep Dive into KABR: A Dataset for Understanding Ungulate Behavior from In-Situ Drone Video**
M. Kholiavchenko, **J. Kline**, M. Kukushkin, O. Brookes, S. Stevens, et al.
Multimedia Tools and Applications 84(21), **2025**.

[J8] **Impact of Drone Disturbances on Wildlife: A Review**
S. Afridi, L. Laporte-Devylder, G. Maalouf, **J. Kline**, S. G. Penny, et al.
Drones 9(4), article 311, **2025**. JIF **5.2**.

Conference and Workshop Articles

[C1] **What the Sheep Can Teach the Shepherd: A Vision for Self-Organizing Drone Swarms Informed by Collective Animal Behavior**
J. Kline, S. Stevens, D. I. Rubenstein, T. Berger-Wolf, and C. Stewart
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS) (to appear), **2026**.
Vision paper.

[C2] **A Maneuver-Indexed Testbed for Context-Aware Adaptive Wildlife Drones**
J. Kline
IEEE ACSOS Companion (ACSOS-C) (to appear), **2026**.
Artifact paper.

[C3] **WildBox: A Dataset and Benchmark for Aerial Monocular 3D Detection of African Savanna Wildlife**
V. Shukla, K. Meier, L. Laporte-Devylder, C. R. Saint-Jean, **J. Kline**, et al.
Manuscript under review, **2026**.
Full dataset paper.

[C4] **Cross-Modal Corroboration for Annotation-Free Wildlife Monitoring**
B. Pillai*, V. Viswapriyan*, C. Stewart, T. Berger-Wolf, and **J. Kline**
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2026**.
Workshop paper.

[C5] **Autonomous UAV Navigation for Individual Wildlife Re-Identification**
C. Sun*, T. Berger-Wolf, and **J. Kline**
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2026**.
Workshop paper.

[C6] **Environment-Aware Dynamic Pruning for Pipelined Edge Inference**
A. O'Quinn, C. Snedeker, S. Zhang, and **J. Kline**
IEEE International Conference on Edge Computing and Communications (EDGE), **2025**.
Full paper. **26%** AR.

[C7] **SmartWilds: Multimodal Wildlife Monitoring Dataset**
J. Kline, A. Potlapally*, B. Pillai*, T. Wani, R. Katole, et al.
Third Workshop on Imageomics at NeurIPS, **2025**.
Workshop paper.

[C8] **Mind the (Data) Gap: Evaluating Vision Systems in Small Data Applications**
S. Stevens, S. M. Rayeed, and **J. Kline**
Third Workshop on Imageomics at NeurIPS, **2025**.
Workshop paper.

[C9] **MMLA: Multi-Environment, Multi-Species, Low-Altitude Aerial Footage Dataset**
J. Kline, S. Stevens, G. Maalouf, C. R. Saint-Jean, D. N. Ngoc, et al.
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2025**.
Workshop paper.

[C10] **WildLive: Near Real-Time Visual Wildlife Tracking Onboard UAVs**
N. N. Dat, T. Richardson, M. Watson, K. Meier, **J. Kline**, et al.
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2025**.
Workshop paper.

[C11] **An Adaptive Autonomous Aerial System for Dynamic Field Animal Ecology Studies**
J. Kline
IEEE ACSOS Companion (ACSOS-C), **2024**.
PhD Forum extended abstract, presented as talk.

[C12] **KABR: In-Situ Dataset for Kenyan Animal Behavior Recognition from Drone Videos**
M. Kholiavchenko, **J. Kline**, M. Ramirez, S. Stevens, A. Sheets, et al.
IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), **2024**.
Workshop paper.

[C13] **Integrating Biological Data into Autonomous Remote Sensing Systems for In Situ Imageomics: A Case Study for Kenyan Animal Behavior Sensing with Unmanned Aerial Vehicles**
J. Kline, M. Kholiavchenko, O. Brookes, T. Berger-Wolf, C. V. Stewart, et al.
First Workshop on Imageomics at AAAI, **2024**.
Workshop paper.

[C14] **Decentralized Multi-Drone Coordination for Wildlife Video Acquisition**
D. Grushchak*, **J. Kline**, D. Pianini, N. Farabegoli, G. Aguzzi, et al.
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), **2024**.
Full paper. **24%** AR. **ACSOS 2024 Best Poster Award.**

[C15] **An Agent-Based Model of Directional Multi-Herds**
D. Grushchak*, **J. Kline**, D. Pianini, and N. Farabegoli
IEEE ACSOS Companion (ACSOS-C), **2024**.
Workshop paper.

Invited Lectures and Presentations *Sole author unless otherwise noted.*

Conservation Technology and Ecology

[P1] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P2] **A Metadata-First FAIR4AI Checklist and Automated Evaluation Agent for Biodiversity Datasets**
E. Sokol et al., including **J. Kline**
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P3] **Cross-Modal Coordination for Biodiversity Monitoring at Scale: Lessons from SmartWilds**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P4] **Digital Twins as System-Optimization Tools: Provisioning Edge and Cloud Infrastructure for Biodiversity Monitoring at Scale**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P5] **kabr-tools: A Modular Workflow for Automated Video-Based Behavioral Monitoring Across the Autonomy Spectrum**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P6] **Lessons from FAIR^2 Drones: Toward AI-Ready Multimodal, Multispatial Sensor Data for Ecology**
Invited symposium talk, "Building an AI-Ready Ecology and Biodiversity Data Infrastructure for Science and Action," Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, **2026**.

[P7] **AI for Ecology: Accelerating Discoveries, Reducing Uncertainties, and Scaling Solutions**
J. Kline, panelist
Special Session SS23, Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, **2026**.

[P8] **Autonomous Drones for Animal Studies with Edge AI**
Invited talk, Global Conservation Tech & Drone Forum, Nairobi, Kenya, **2026**.

[P9] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline et al.
Poster presentation, FARR Workshop: FAIR in ML, AI Readiness and Reproducibility, Washington, DC, **2026**.

[P10] **How Do Drones Fit into Multimodal Sensing Networks?**
Poster presentation, WildDrone Summer School, Max Planck Institute of Animal Behavior, Konstanz, Germany, **2025**.

[P11] **Autonomous Drones for Collective Wildlife Behavior**
Invited lecture, WildDrone Seminar Series, **2025**.

Edge AI and Autonomous Systems

[P12] **An Edge-Native Approach to Behavior-Adaptive Navigation in Drone Systems**
J. Kline, R. Katole, and C. Stewart
Refereed extended abstract, SRC poster, ACM/IEEE Symposium on Edge Computing (SEC), **2025**.
SEC 2025 Student Research Competition — First Place.

[P13] **An Edge-to-Cloud Framework for Vigilance-Adaptive Drones**
P. Covey*, **J. Kline**, and C. Stewart
Refereed extended abstract, SRC poster, ACM/IEEE Symposium on Edge Computing (SEC), **2025**.

[P14] **Autonomous UAV Missions for Studying Wildlife Behavior: A Case Study for the Individual Identification of Zebras**
Poster presentation, Midwest Machine Learning Symposium (MMLS), Chicago, IL, **2023**.

[P15] **Seeing the Trees for the Forest: Describing HPC Filesystem Trees with the Grand Unified File-Index (GUFI)**
J. Kline, J. Lee, and R. Davis
Refereed extended abstract, poster presentation, ACM Student Research Competition (SRC), International Conference for High Performance Computing, Networking, Storage, and Analysis (SC23), Denver, CO, **2023**.

[P16] **Edge Computing for Software-Defined Cartography**
PhD Forum extended abstract, presented as talk, ACM/IEEE Symposium on Edge Computing (SEC), Seattle, WA, **2022**.

Invited University Lectures and Seminars

[P17] **Drones and AI in Field Animal Ecology: Integrating Biology, Animal Science, and Technology**
Invited lecture, Fridays at Findlay, University of Findlay, **2025**.

[P18] **From Ohio to Kenya: Autonomous Drones in Field Ecology and Conservation**
Invited lecture, Conservation Science Symposium, Muskingum University and The Wilds Conservation Center, **2025**.

[P19] **Adaptive, Autonomous Aerial Systems for Dynamic Field Animal Ecology Studies**
Invited lecture, University of Bologna, Cesena, Italy, **2024**.

Institutional and Broadening-Participation Venues

[P20] **Autonomous AI-Driven Environmental Sensing Systems**
Poster presentation, MIT and Boston University Rising Stars in EECS Workshop, **2025**.

[P21] **Autonomous, Adaptive Vision-Based Remote Sensing System for Dynamic Field Animal Ecology Studies**
Oral presentation, Edward F. Hayes Advanced Research Forum, The Ohio State University, **2025**.
First Place, Engineering Oral Presentation.

[P22] **Individual Identification of Zebras with Autonomous UAV Swarms**
Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, **2024**.
Best Poster Award.

[P23] **Intelligent Cyberinfrastructure for Remote Sensing with Autonomous Aerial Robotics**
Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, **2023**.

[P24] **Individual Identification of Zebras with Autonomous UAV Swarms**
Poster presentation, CRA-WP Grad Cohort for Women, San Francisco, CA, **2023**.

[P25] **Interdisciplinary Applications of Autonomous Unmanned Aerial Vehicles**
Poster presentation, STARS Celebration at Tapia and Ohio Celebration of Women in Computing, **2023**.

Artifacts: Models, Datasets, and Software

[A1] **FAIR^2 Drones: Data Infrastructure for AI-Ready Wildlife Drone Datasets**
Software and datasets: GitHub, Hugging Face. *FAIR, AI-ready, Darwin Core–compliant data infrastructure for multi-modal animal ecology datasets. Described in [F1, P1, P9].*

[A2] **kabr-tools: Automated Framework for Multi-Species Behavioral Monitoring**
Software: GitHub. *Open-source pipeline for analyzing animal behavior from in situ video. Used in [J1, P5].*

[A3] **A Maneuver-Indexed Evaluation Framework for Context-Aware Adaptive Wildlife Drones**
Software and data: Hugging Face Spaces. *Interactive research demo and evaluation data. Companion to [C2].*

[A4] **WildWing: Open-Source Autonomous UAS for Animal Behaviour Monitoring**
Software: GitHub. *Open-source drone system for autonomous animal monitoring. Companion to [F5].*

[A5] **In Situ Kenyan Animal Behavior Recognition (KABR) Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* data collection, annotation, analysis, telemetry integration, and public release. *Drone videos, annotations, synchronized telemetry, and models. Used in [J1, J4, J7, C12, C15].*
141K downloads

[A6] **Multi-Environment, Multi-Species, Low-Altitude Aerial (MMLA) Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* data collection, annotation, analysis, curation, and public release. *Multi-site detection and pose datasets, and detection and pose models trained on them. Used in [C2, C9].*
199K downloads

[A7] **SmartWilds: Multimodal Wildlife Monitoring Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* project lead; led study design, field data collection, annotation, analysis, multimodal coordination, and public release. *Camera-trap, bioacoustic, and drone datasets. Used in [F2, C4, C7].*
12K downloads

References

Dr. Tanya Berger-Wolf — *PhD co-advisor*
Director, Translational Data Analytics Institute; Professor of Computer Science and Engineering, Electrical and Computer Engineering, and Evolution, Ecology, and Organismal Biology, The Ohio State University
Director, NSF Imageomics Institute and the US–Canada AI and Biodiversity Change (ABC) Global Climate Center
berger-wolf.1@osu.edu

Dr. Christopher Stewart — *PhD co-advisor*
Professor, Computer Science and Engineering, The Ohio State University
stewart.962@osu.edu

Dr. Daniel I. Rubenstein — *Ecology mentor and collaborator*
Class of 1877 Professor of Zoology, Emeritus, and Senior Scholar, Ecology and Evolutionary Biology, Princeton University
dir@princeton.edu

Dr. Tilo Burghardt — *PhD committee member*
Professor of Computer Vision and Animal Biometrics, University of Bristol, UK; Principal Investigator, WildDrone
tilo@cs.bris.ac.uk

Last updated: August 7, 2026